Comprehensive Analysis & Drug Target Identification

HRA001684 — Hashimoto's Thyroiditis Bulk RNA-seq

Zhang et al. 2022, *Nature Communications*

---

1. Dataset Overview

Parameter	Value
Dataset	HRA001684 (NGDC GSA-Human)
Publication	Zhang et al. 2022, *Nat Commun* (DOI: 10.1038/s41467-022-28120-2)
Samples	66 thyroid tissue biopsies (16 HT, 50 Non-HT), 3 batches
Expression data	DESeq2 size-factor normalized counts (authors' MOESM7)
DE method	Authors' LME + DESeq2 (batch-corrected)
Genes measured	28,029

Important data limitation: Raw FASTQ files were not re-processed; all DE statistics are from the authors' published supplementary results. Expression values are size-factor normalized (not raw counts), which is appropriate for visualization and correlation but not for re-running DESeq2.

---

2. Differential Expression Landscape

2.1 Summary Statistics

- **2,242 significant DEGs** (padj < 0.05, |log2FC| ≥ 1)
- **2,107 upregulated** vs **135 downregulated** — a striking **15.6× asymmetry**
- Median |log2FC|: 2.03 (up), 1.20 (down)

- Maximum upregulation: TERT (log2FC = 9.83), SERPINA9 (9.18), FCAMR (8.32)
- Maximum downregulation: MYOC (−2.19), AMER3 (−2.19), GLDC (−2.03)

The extreme up/down asymmetry is **biologically expected**: HT is a gain-of-function immune infiltration disease. The upregulated genes reflect lymphocyte invasion; the downregulated genes (MYOC, GLIS3-AS1, GLDC) are thyroid follicular cell markers being displaced by infiltrating immune cells.

2.2 Gene Family Breakdown

Family	Up	Down	Total
Cytokines/Chemokines	80	1	81
CD Antigens	72	0	72
Thyroid/Stromal	42	11	53
NK/Lymphocyte Receptors	24	0	24
Innate Immune	20	0	20
HLA	19	0	19
B-cell Program	10	0	10
T-cell Program	9	0	9

2.3 B-cell Marker Co-expression

All 15 queried B-cell markers (PAX5, FCRL1/4, CR2, TCL1A, SPIC, AICDA, CD19, CD79A/B, BTK, BANK1, MS4A1, CD22, TNFRSF17) are uniformly upregulated in HT samples, consistent with the massive B-cell infiltration that defines Hashimoto's thyroiditis histopathology.

---

3. Pathway Enrichment Analysis

3.1 GSEA Results (1,006 significant gene sets)

Category	Activated	Suppressed
GO:BP	884	51
KEGG	44	9
Hallmark	15	3

Note: 18 Hallmark gene sets reached significance (15 activated, 3 suppressed).

Top activated Hallmark pathways:

Pathway	NES	padj
Allograft Rejection	2.16	2.7×10^{-9}
IFN-γ Response	2.04	2.7×10^{-9}
IFN-α Response	2.00	2.7×10^{-9}
IL6-JAK-STAT3	1.98	2.7×10^{-9}
Inflammatory Response	1.93	2.7×10^{-9}

Suppressed Hallmark pathways: Oxidative Phosphorylation (NES = −1.59), Adipogenesis (−1.36), Myogenesis (−1.19) — reflecting displacement of metabolically active thyrocytes.

3.2 Pathway Clustering

Hierarchical clustering of Hallmark pathways by leading-edge Jaccard similarity reveals 4 clusters:

1. **Immune activation** (n=10, mean NES=1.84): allograft rejection, IFN responses, IL6-JAK-STAT3, TNFα/NF-κB, complement, apoptosis
2. **Cell cycle** (n=5, mean NES=1.61): E2F targets, G2M checkpoint, MYC targets, mTORC1 — likely reflecting proliferating lymphocytes
3. **Metabolic suppression** (n=2, mean NES=−1.47): oxidative phosphorylation, adipogenesis
4. **Myogenesis** (n=1, NES=−1.19)

Note on Jaccard similarity: Mean within-cluster Jaccard = 0.079 vs overall mean = 0.031. The clustering is statistically meaningful but modest — Hallmark gene sets are designed to be non-redundant, so low overlap is expected. The clusters are primarily driven by NES sign (immune activation vs metabolic suppression) rather than gene-level overlap.

3.3 Pathway-Central Genes

Genes appearing in the most GSEA leading edges + ORA significant pathways (pathway centrality score):

Rank	Gene	Centrality	log2FC	Biological role
1	TNF	443	2.53	Master inflammatory cytokine
2	TGFB1	428	1.30	Immune regulation, fibrosis
3	LGALS9	398	1.83	TIM-3 ligand, immune checkpoint
4	PTPRC	362	4.16	CD45, pan-leukocyte marker
5	TREM2	341	2.19	Macrophage activation
6	IL12B	329	3.07	IL-12/23 p40 subunit, Th1 driver

---

4. Drug Target Prioritization

4.1 Scoring Framework

46 top pathway-central DEGs were queried against the Open Targets Platform (v26.03) and scored on 5 dimensions:

Dimension	Weight	Rationale
Druggability (SM/AB tractability + clinical stage)	30%	Feasibility of therapeutic intervention
Pathway centrality	25%	Network importance in HT biology
Expression magnitude (	log2FC	)

Dimension	Weight	Rationale
Disease relevance (OT association)	15%	Evidence linking target to thyroid autoimmunity
Autoimmune precedent (approved autoimmune drugs)	10%	Safety profile and translational readiness

4.2 Ranking Results

Rank	Gene	Score	Approved Drug	HT Association
1	IFNG	0.827	Emapalumab	Hypothyroidism (OT)
2	**IL12B**	**0.814**	**Ustekinumab**	**Autoimmune thyroid disease (OT)**
3	TNF	0.781	Etanercept/Adalimumab	None (top 50)
4	BTK	0.754	Ibrutinib/Rilzabrutinib	None (top 50)
5	JAK3	0.697	Ritlecitinib/Tofacitinib	None (top 50)

4.3 Critical Caveats on Rankings

IFNG (rank 1) is overranked due to a scoring artifact: its disease score is inflated by an Open Targets "hypothyroidism" association (score=0.434), but hypothyroidism is the *downstream outcome* of HT, not the autoimmune process itself. Furthermore, emapalumab is approved for hemophagocytic lymphohistiocytosis (HLH), a rare pediatric disease unrelated to HT. Blocking IFN-γ systemically carries serious infection risk given its essential role in anti-viral and anti-bacterial immunity.

TNF (rank 3) carries a documented safety contraindication: anti-TNF therapy (adalimumab, infliximab) can paradoxically *induce* destructive thyroiditis and autoimmune thyroid disorders as adverse events. Multiple case reports and reviews confirm this paradoxical effect. TNF inhibition is therefore contraindicated specifically for HT.

BTK (rank 4) is mechanistically underranked: BTK is the essential kinase for B-cell receptor signaling, and HT is defined by B-cell infiltration and autoantibody production (anti-TPO, anti-Tg). BTK inhibitors (rilzabrutinib) are already approved for autoimmune

diseases. The lack of a direct HT Open Targets association penalizes BTK in the scoring, but its mechanistic rationale is very strong.

---

5. Top Drug Target: IL12B

5.1 Rationale for Selection

IL12B (encoding the p40 subunit shared by IL-12 and IL-23) is selected as the primary drug target recommendation based on convergent evidence across all analytical layers:

1. **Strong differential expression:** log2FC = +3.07, padj = 3.4×10^{-9}, ~8× higher in HT vs Non-HT thyroid tissue
2. **High pathway centrality:** rank 6 of 2,242 DEGs (centrality score = 329), appearing in 161 GSEA leading edges and 168 ORA significant pathways
3. **Direct disease association:** Open Targets lists "autoimmune thyroid disease" as an associated disease (score = 0.204)
4. **Genetic evidence:** IL12B SNPs are associated with HT clinical presentation and severity [11][12][16]
5. **Elevated serum IL-12p40 in HT patients** confirmed by independent ELISA studies [12]
6. **Approved drug with autoimmune precedent:** Ustekinumab (anti-IL12/23 p40 monoclonal antibody) is approved for psoriasis, psoriatic arthritis, Crohn's disease, and ulcerative colitis — all Th1/Th17-driven autoimmune diseases with mechanistic parallels to HT
7. **No safety concern of worsening thyroid autoimmunity** (unlike TNF inhibitors)

5.2 Mechanism of Action in HT

```
Thyroid-infiltrating macrophages/DCs
         ↓ produce
    IL-12 (p35 + p40)  ←── IL12B encodes p40
         ↓ signals via IL-12Rβ1/β2
    STAT4 phosphorylation
         ↓
    Th1 differentiation (TBX21/T-bet)
         ↓ produces
```

```
IFN-γ (IFNG, r=0.72 within HT)
     ↓
CXCL9/CXCL10 production (r=0.84/0.83 with IL12B)
     ↓ recruits
CD8+ cytotoxic T cells (r=0.83 with IL12B)
     ↓
Thyrocyte destruction + anti-TPO/anti-Tg antibody production
```

IL-12 also shares the p40 subunit with IL-23, which drives Th17 differentiation. Serum IL-23 is elevated in HT [24], and ustekinumab blocks both IL-12 and IL-23 simultaneously — a dual mechanism particularly relevant given the Th1/Th17 mixed phenotype of HT.

5.3 Co-expression Evidence

Within HT samples (n=16), IL12B is significantly co-expressed with:

Gene	Within-HT r	p-value	Biological link
CXCL9	0.78	<0.001	IFN-γ-induced chemokine, T-cell recruitment
GZMB	0.75	0.001	Cytotoxic T-cell effector
CD8A	0.68	0.004	Cytotoxic T-cell marker
CXCL10	0.76	<0.001	IFN-γ-induced chemokine
IFNG	0.72	0.002	Th1 effector cytokine
PRF1	0.66	0.006	Perforin, cytotoxic killing

Important caveat: Overall correlations (n=66) are partly driven by group separation (HT vs Non-HT). Within-group correlations (reported above) are the biologically meaningful values. The IL12B–BTK overall correlation (r=0.61) is not maintained within HT (r=0.01), indicating it reflects group-level co-upregulation rather than true co-regulation.

5.4 Therapeutic Opportunity

Aspect	Assessment
Drug class	Anti-IL12/23 p40 monoclonal antibody
Lead compound	Ustekinumab (Stelara, Janssen)

Aspect	Assessment
Mechanism	Blocks IL-12 and IL-23 simultaneously
Autoimmune approvals	Psoriasis, PsA, Crohn's disease, UC
HT clinical trials	None completed — **novel repositioning opportunity**
Safety profile	Well-characterized; infection risk manageable
Biomarker	Serum IL-12p40, anti-TPO antibody titer

5.5 Secondary Candidate: BTK

BTK (rank 4, log2FC=4.39, pathway centrality=263) is the strongest mechanistic secondary candidate. HT is fundamentally a B-cell-mediated disease (anti-TPO, anti-Tg autoantibodies), and BTK is the essential kinase for B-cell receptor signaling and B-cell survival. BTK inhibitors (rilzabrutinib) are already approved for autoimmune thrombocytopenic purpura and pemphigus, establishing autoimmune precedent. The lack of a direct HT Open Targets association reflects a data gap rather than a biological argument against BTK.

---

6. Key Limitations

1. **Expression data source:** All analyses use the authors' published supplementary data (DESeq2 normalized counts + LME-corrected DE statistics), not re-processed raw reads. Results cannot be independently validated without raw FASTQ access.
2. **Sample size:** 16 HT vs 50 Non-HT (3.1× imbalance). Within-HT correlations (n=16) have limited statistical power; treat within-HT r values as exploratory.
3. **Bulk RNA-seq limitation:** Expression reflects the average of all cell types in the thyroid biopsy. The massive immune infiltration in HT means many "upregulated" genes reflect cell-type composition changes (more lymphocytes) rather than transcriptional changes within a given cell type. Single-cell RNA-seq (Zhang et al. 2022 also published scRNA-seq data) would be needed to resolve cell-type-specific expression.
4. **Pathway centrality metric:** The centrality score (GSEA leading-edge + ORA pathway

count) is a proxy for biological connectivity, not a direct measure of causal importance. Highly connected genes (TNF, TGFB1) may be downstream effectors rather than upstream drivers.

5. **Drug target scoring:** The composite scoring framework uses subjective weights. IFNG's overranking demonstrates that disease association scores from Open Targets can conflate related but distinct phenotypes (hypothyroidism ≠ Hashimoto's autoimmune process). Scores should be interpreted as a prioritization aid, not a definitive ranking.
6. **No functional validation:** All findings are computational. Experimental validation (e.g., IL-12 blockade in a murine HT model, or analysis of ustekinumab-treated patients with comorbid HT) would be required before clinical translation.

---

7. Figures Generated

Figure	Description
`PCA_plot.png`	PCA of 66 samples (PC1=46.5%, PC2=6.9%)
`Volcano_plot.png`	Volcano plot of 18,983 genes
`MA_plot.png`	MA plot
`Heatmap_top50_DEGs.png`	Heatmap of top 50 DEGs
`DEG_landscape_analysis.png`	4-panel DEG landscape (LFC distribution, gene families, pathway centrality scatter, B-cell heatmap)
`GSEA_dotplot.png`	GSEA dotplot (top 20 pathways)
`GSEA_running_score.png`	GSEA running score plots
`ORA_upregulated_barplot.png`	ORA barplot (upregulated DEGs)
`pathway_jaccard_heatmap.png`	Hallmark pathway Jaccard similarity heatmap
`pathway_NES_barplot.png`	Hallmark NES barplot by cluster
`leading_edge_genes_bubble.png`	Top 25 leading-edge genes bubble chart
`drug_target_prioritization.png`	Drug target ranking (stacked bar + summary table)

Figure	Description
`drug_target_ranking.csv`	Full ranked table of 46 candidates with scores
`IL12B_deep_dive.png`	IL12B expression, co-expression, and target profile

---

8. Methods Summary

- **Differential expression:** Authors' LME + DESeq2 (batch-corrected, 3 batches)
- **Significance threshold:** padj < 0.05, |log2FC| ≥ 1
- **GSEA:** fgsea via clusterProfiler; Hallmark + KEGG_LEGACY + GO:BP gene sets (MSigDB)
- **ORA:** clusterProfiler enricher; upregulated DEGs vs background
- **Pathway centrality:** Sum of GSEA leading-edge appearances + ORA significant pathway appearances per gene
- **Drug target scoring:** 5-dimension weighted composite (druggability 30%, centrality 25%, LFC 20%, disease relevance 15%, autoimmune precedent 10%)
- **Open Targets query:** GraphQL API v26.03 (April 2026); `drugAndClinicalCandidates`, `tractability`, `associatedDiseases` fields
- **Correlations:** Pearson r; reported both overall (n=66) and within-group (HT n=16, Non-HT n=50)

---

9. References

Key references supporting the drug target recommendation:

- Zhang et al. 2022, *Nat Commun* — source dataset (HRA001684)
- Walsh et al. 2011, *Clin Endocrinol* — IL12B genotype and HT clinical presentation [11]
- Gerenova et al. 2016, *Biotechnol Biotechnol Equip* — IL-12p40 elevated in HT serum [12]
- Inoue et al. 2017, *Endocr J* — IL12 polymorphisms and HT severity [16]
- Ruggeri et al. 2014, *Endocr J* — IL-23 elevated in HT [24]
- Kumagai et al. 2026, *Front Immunol* — anti-TNF-induced destructive thyroiditis [1]

- Open Targets Platform v26.03 — druggability and disease association data